DATE:-14/08/2025

TASK-2

Generating design of other traditional database model creating hierarchical/network model of the database by enhancing the sound abstract data by performing following tasks using forms of inheritance:

2.a. Identify the specificity of each relationship, find and form surplus relations.

2.b. check is-a hierarchy / has-a hierarchy and performs generalization and/or specialization relationship.

2.c. Find the domain of the attribute and perform a check constraint to the applicable

2.d. Rename the relations.

2.e. Perform SQL Relations using DDL, DCL commands.

IMPLEMENTATION OF DDL, DML, DCL and TCL COMMANDS OF SQL

AIM:- Implementation of DDL, DML, DCL and TCL commands of SQL with suitable examples

Objective:

* To understand the different issues involved in the design and implementation of a database system.
* To understand and use data definition language to write query for a database

Theory:

Oracle has many tools such as SQL*PLUS, Oracle Forms, Oracle Report Writer, Oracle Graphics etc.

SQL*PLUS: The SQL * PLUS tool is made up of two distinct parts. These are

Interactive SQL: Interactive SQL is designed for create, access and manipulate data structures like tables and indexes.

PL/SQL: PL/SQL can be used to developed programs for different applications.

Oracle Forms: This tool allows you to create a data entry screen along with the suitable menu objects. Thus it is the oracle forms tool that handles data gathering and data validation in a commercial application.

Report Writer: Report writer allows programmers to prepare innovative reports using data from the oracle structure like tables, views etc. It is the report writer tool that handles the reporting section of commercial application.

Oracle Graphics:- some of the data can be better represented in the form of pictures. The oracle graphics tool allows programmers to prepare graphs using data from oracle structures like tables, views etc.

SQL (Structured Query Language):

Structured Query Language is a database computer language designed for managing data in relational database management systems (RDBMS), and originally based upon Relational Algebra. Its scope includes data query and update, schema creation and modification, and data access control

SQL was one of the first languages for Edgar F. codd's relational model and became the most widely used language for relational database.

IBM developed SQL in mid of 1970's

Oracle incorporated in the year 1979.

SQL used by IBM/DB2 and DS database systems.

SQL adopted as standard language for RDBS by ASNI in 1989.

DATA TYPES:

1. CHAR (size): This data type is used to store character strings values of fixed length. The size in brackets determines the number of characters the cell can hold. The maximum number of character is 255 characters.

2. VARCHAR (size)/ VARCHAR2 (size): This data type is used to store number (fixed or floating point). Number of virtually any magnitude may be stored up to 38 digits of precision

3. NUMBER (p,s): The NUMBER data type is used to store number (fixed or floating point). Number of virtually any magnitude may be stored up to 38 digits of precision.

Number as large as $9.99 * 10^{124}$. The precision (p) determine the number of places to the right of the decimal. If scale is omitted then the default is zero. If precision is omited, values are stored with their original precision up to the maximum of 38 digits.

4. DATE:- This data types is used to represent date and time. The standard format is DD-MM-YY as in 17-SEP-2009. TO enter dates other than the standard format, use the

appropriate functions. Date time stores date in the 24-hours format By default the time in a date field is 12:00:00 am, if no time portion is specified. The default date for a date field is the first day the current month.

5- LONG: This data type is used to store variable length character strings containing up to 2GB. Long data can be used to store arrays of binary data in ASCII format LONG values cannot be indexed, and the normal character functions such as SUBSTR cannot be applied.

6. RAW:- The RAW data type is used to store binary data, such as digitized picture or image. Data loaded into columns of these data types are stored without any further conversion. RAW data types can have a maximum length of 255 bytes. LONG RAW data type can contain up to 2GB.

DATA DEFINITION LANGUAGES: (For Shopping Management System)

A DATA Definition Language (DDL) statement are used to define the database structure or schema.

CREATE:- To make a new database, table, index, or stored query. A create statement in SQL creates an object inside of a relational database management system (RDBMS)

SYNTAX:-

CREATE TABLE table_name

(column_name1 data-type (Esize]), column_name2 data_type (Esize])...

column_name_n data-type (Esize]));

ALTER:- To modify an existing database object. Alter the structure of the database

- To add a column in a table

SYNTAX

ALTER TABLE table_name ADD column_name datatype([size]);

- To delete a column in a table

SYNTAX

ALTER TABLE table_name DROP column column_name;

- To modify a column in a table

SYNTAX:-

ALTER TABLE table_name modify column_name data_type ([new_size]);

RENAME

SYNTAX:

ALTER TABLE table_name RENAME column old_column_name to new_column_name;

DESCRIBE

SYNTAX

DESC table_name

TRUNCATE TABLE:

Remove all records from a table, including all spaces allocated for the records are removed

SYNTAX:

TRUNCATE TABLE table_name

DATA MANIPULATION LANGUAGES:

Data manipulation language allows the users to query and manipulate data in existing schema in object. It allows following data to insert, delete, update and recovery data in schema object

INSERT:- Values can be inserted into table using insert commands. There are two types of insert commands - They are multiple value insert commands (using '&' symbol) single value insert command (without using '&' symbol)

SYNTAX

```
INSERT INTO table_name VALUES (value1, value2, value3, ...);
(OR)
INSERT INTO table_name (column1, column2, column3, ...) VALUES
(value 1, value 2, value 3, ....);
```

UPDATE:- This allows the user to update the particular column value using the where clause condition.

SYNTAX

```
UPDATE <table_name> SET <COL1 = value> WHERE <column=value>;
```

DELETE:- This allows you to delete the particular column values using where clause condition

SYNTAX:

```
DELETE FROM <table_name> WHERE <conditions>;
```

SELECT:- The select statement is used to query a database. This statement is used to retrieve the information from the database. The SELECT statement can be used in many ways - They are

1. Selecting Some columns:

To select specified number of columns from the table the following command is used

SYNTAX

SELECT column_name FROM table_name;

2. Select using DISTINCT:-

The DISTINCT keyword is used to return only different values (i.e.) this command does not select the duplicate values from the table

SYNTAX

SELECT DISTINCT column name(s) FROM table_name;

3. Select using BETWEEN

BETWEEN can be used to get those items that fall within a range.

SYNTAX

SELECT column name FROM table_name WHERE column name BETWEEN value1 AND value2;

4. Renaming:

The Sclect statement can be used to rename either a column or the entire table.

SYNTAX

Renaming a column:

SELECT column name AS new name FROM table_name;

5. SORTING

The select statement with the order by clause is used to sort the contents

Table either in asscending or descending order.

SYNTAX

SELECT column name FROM table_name WHERE condition ORDER BY column name ASC/DESC;

6. To select by matching some patterns:

The select statement along with like clause is used to match strings. The like condition is used to specify a search pattern in a column.

SYNTAX:-

SELECT column name FROM table_name WHERE column name LIKE "%o or _";

% : Matches any sub string

_ : Matches a single character.

7. Select using AND, OR, NOT:

We can combine one or more conditions in a SELECT statement using the logical operators AND, OR, NOT.

SYNTAX

SELECT column name FROM table_name WHERE condition1 LOGICAL OPERATOR condition2

DATA CONTROL LANGUAGES:

1. Create:

• create user kamal identified by kunal;

User created.

2- Grant:

• grant all privileges to kamal;

Grant succeeded.

3. Revoke:

• revoke all privileges from kamal;

Revoke succeeded.

TRANSACTION CONTROL LANGUAGES

1. commit:

```
• commit;
commit complete.
```

2. save point:

```
• savepoint k1;
Savepoint created
```

3. Rollback:

```
• rollback to k1;
Rollback complete
```

OUTPUT CREATE TABLE table_name (column_name1 data_type ([size]), column_name2 data_type ([size]) ... column_name_n data_type ([size]);

```
SQL> desc icoli
```

Name	NULL?	TYPE
NAME		VARCHAR2(12)
VTUNO		NUMBER(5)
ADDRESS		VARCHAR2(14)

```
SQL> desc icoli
```

o/p

Name	NULL?	TYPE
NAME		
VTUNO		
ADDRESS		

ALTER TABLE table_name ADD column_name datatype([size])

```
SQL> desc icoli (to add column)
```

Name	NULL?	Type
NAME		VARCHAR(12)
VTUNO		NUMBER(5)
ADDRESS		VARCHAR(14)
PHONENO		NUMBER(10)

ALTER TABLE table_name DROP column (column_name)

```
SQL> desc icoli (to drop colum)
```

Name	NULL?	Type
NAME		VARCHAR(12)
VTUNO		NUMBER(5)
PHONENO		NUMBER(10)

ALTER TABLE table_name modify column_name data_type.
[(new-size)];

SQL> desc icoh (to modify column)

Name	NULL?	TYPE
NAME		VARCHAR(12)
VTUNO		NUMBER(3)
PHONENO		NUMBER(10)

ALTER TABLE table_name Rename column ~~modify~~ old_column_name ~~data_type~~ to new_column_name;

SQL> desc icoh (to rename column)

Name	NULL?	TYPE
STUDENT_NAME		VARCHAR(12)
VTUNO		NUMBER(3)
PHONENO		NUMBER(10)

INSERT INTO table_name VALUES (value1, value2, ...);

SQL> desc icoh (to insert value)

~~Name~~	VTUNO	PHONENO
STUDENT_NAME		
Shivapriya	299	9515221655
mahalakshmi	275	9876543210
Bhavyasree	284	9768543210

SQL> UPDATE icoh set student_name = 'shivapriya' WHERE VTUNO=289;

STUDENT_NAME	VTUNO	PHONENO
shivapriya	289	9515221655
mahalakshmi	275	9876543210
Bhavyasree	284	9768543210

SQL> DELETE FROM icoh WHERE STUDENT_NAME = Bhavyasree;

STUDENT_NAME	VTUNO	PHONENO
shivapriya	289	9515221655
mahalakshmi	275	9876543210

RESULT:- Thus, The implementation of DDL, DML, DCL and TCL commands of SQL with suitable example

VEL TECH	
S. NO.	2
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

SQL> SELECT VTUNO FROM KOTI;

VTUNO
289
275

SQL> SELECT DISTINCT column PHONENO FROM KOTI;

PHONENO
9515221655
9876543210

SQL>

SELECT * FROM KOTI WHERE VTUNO BETWEEN 280 and 290;

VTUNO
289
~~285~~

SQL> SIELECT STUDENT_NAME AS People_name FROM KOTI;

People_name	VTUNO	PHONENO
Shivapriya	289	9515221655
Manalaleshmi	275	9876543210

SQL> Create user koh identified by kumal;

User created

SQL> grant all privileges to kohj

Grant Succeeded

SQL> Revokej revoke all privileges from kohj

Revoke Succeeded

SQL> Commitj

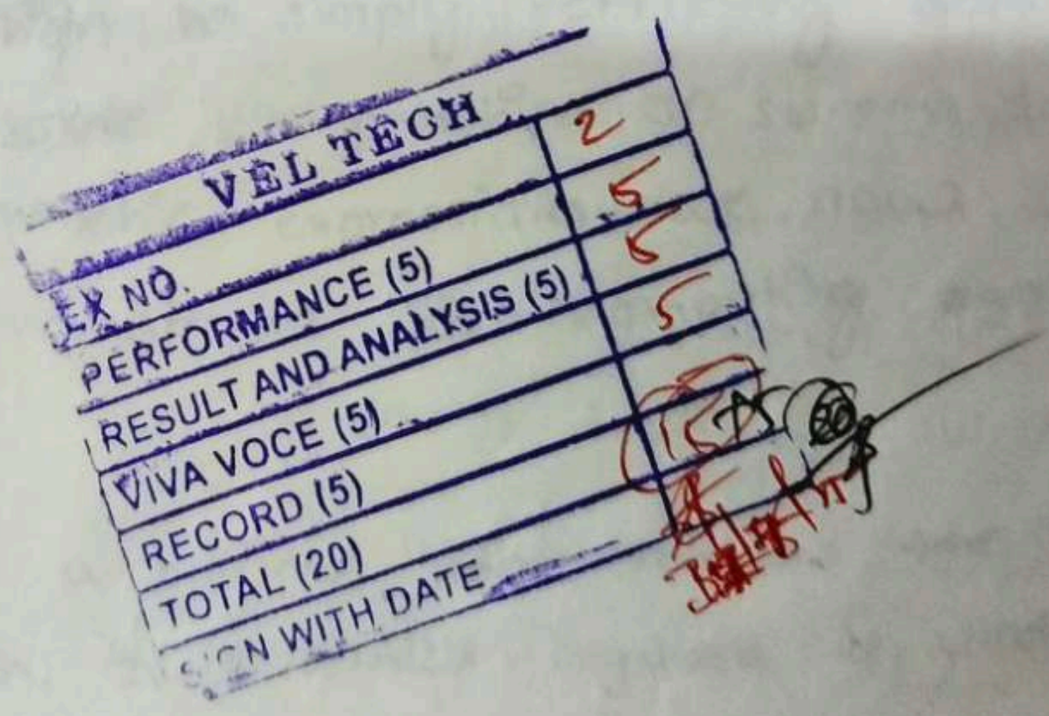
Commit completed

SQL> Savepoint slj

Save point created

SQL> Rollbaele to slj

Roll back complete.



RESULT:- Thus, the Implementation of DDL, DML and TCL Commands in SQL with suitable examples

SQL> connect

Enter user-name: system

Enter password:

Connected.

SQL> create table koteswarao(friends_name varchar(15),phoneno number(10),address varchar(20),salary number(5))

2 ;

Table created.

SQL> desc koteswarao

Name	Null?	Type
FRIENDS_NAME		VARCHAR2(15)
PHONENO		NUMBER(10)
ADDRESS		VARCHAR2(20)
SALARY		NUMBER(5)

SQL> alter table koteswarao add email varchar(15);

Table altered.

SQL> desc koteswarao

Name	Null?	Type
FRIENDS_NAME		VARCHAR2(15)

PHONENO	NUMBER(10)
ADDRESS	VARCHAR2(20)
SALARY	NUMBER(5)
EMAIL	VARCHAR2(15)

SQL> alter table koteswarao drop column email;

Table altered.

SQL> desc koteswarao

Name	Null?	Type
FRIENDS_NAME		VARCHAR2(15)
PHONENO		NUMBER(10)
ADDRESS		VARCHAR2(20)
SALARY		NUMBER(5)

SQL> alter table koteswarao modify salary number(6);

Table altered.

SQL> desc koteswarao

Name	Null?	Type
FRIENDS_NAME		VARCHAR2(15)
PHONENO		NUMBER(10)

```
ADDRESS                        VARCHAR2(20)
SALARY                         NUMBER(6)
```

```
SQL> alter table koteswarao rename column friends_name to best_friends_name;

Table altered.

SQL> desc koteswarao
 Name                          Null?    Type
 ----------------------------- -------- ------------------------
 BEST_FRIENDS_NAME                      VARCHAR2(15)
 PHONENO                                NUMBER(10)
 ADDRESS                                VARCHAR2(20)
 SALARY                                 NUMBER(6)

SQL> insert into koteswarao values('manikanta',8688598110,'vinukonda',180000);

1 row created.

SQL> insert into koteswarao values('anil_kumar',9440774380,'vinukonda',140000);

1 row created.

SQL> insert into koteswarao values('chandrajith',7358959739,'namakal',900000);

1 row created.
```

SQL> insert into koteswarao values('shankar-bhai',8074862610,'kadapa',600000);

1 row created.

SQL> insert into koteswarao values('nithish_kumar',9347734009,'guntur',800000);

1 row created.

SQL> select * from koteswarao;

BEST_FRIENDS_NA PHONENO ADDRESS SALARY

---

manikanta 8688598110 vinukonda 180000

anil_kumar 9440774380 vinukonda 140000

chandrajith 7358959739 namakal 900000

shankar-bhai 8074862610 kadapa 600000

nithish_kumar 9347734009 guntur 800000SQL> update koteswarao set salary=190000 where best_friends_name='manikanta';

1 row updated.

SQL> select * from koteswarao;

BEST_FRIENDS_NA PHONENO ADDRESS SALARY

---

manikanta 8688598110 vinukonda 190000

anil_kumar 9440774380 vinukonda 140000

chandrajith 7358959739 namakal 900000

shankar-bhai 8074862610 kadapa 600000

nithish_kumar 9347734009 guntur 800000

SQL> delete from koteswarao where best_friends_name='anil_kumar';

1 row deleted.

SQL> select * from koteswarao;

BEST_FRIENDS_NA PHONENO ADDRESS SALARY

---------- --------- ---------- ---------

manikanta 8688598110 vinukonda 190000

chandrajith 7358959739 namakal 900000

shankar-bhai 8074862610 kadapa 600000

nithish_kumar 9347734009 guntur 800000

SQL> select best_friends_name from koteswarao;

BEST_FRIENDS_NA

----

manikanta

chandrajith

shankar-bhai

nithish_kumar

SQL> select distnict address from koteswarao;

select distnict address from koteswarao

*

ERROR at line 1:

ORA-00904: "DISTNICT": invalid identifier

SQL> select distinct address from koteswarao;

ADDRESS

---

vinukonda

guntur

kadapa

namakal

SQL> select * from koteswarao where salary between 150000 and 999999;

BEST_FRIENDS_NA	PHONENO	ADDRESS	SALARY
manikanta	8688598110	vinukonda	190000
chandrajith	7358959739	namakal	900000
shankar-bhai	8074862610	kadapa	600000
nithish_kumar	9347734009	guntur	800000

SQL> select address as native_place from koteswarao;

NATIVE_PLACE

---

vinukonda

namakal

kadapa

guntur

SQL> select phoneno from koteswarao order by salary asc;

PHONENO

8688598110

8074862610

9347734009

7358959739

SQL> select salary from koteswarao where best-friends_name like'_0%';

select salary from koteswarao where best-friends_name like'_0%'

*

ERROR at line 1:

ORA-00904: "FRIENDS_NAME": invalid identifier

SQL> select salary from koteswarao where best_friends_name like'_0%';

no rows selected

SQL> select best_friends_name from koteswarao where salary=190000 and phoneno=8688598110;

BEST_FRIENDS_NA

---

manikanta

SQL> create user koteswarao identified by kunal;

User created.

SQL> grant all privileges to koteswarao;

Grant succeeded.

SQL> revoke all privileges from koteswarao;

Revoke succeeded.

SQL> commit;

Commit complete.